

Lab 08: Orchestration

Agenda

Table of Contents

- 01 Introduction
- 02 Apache Airflow
- 03 Conclusion

What is orchestration?

- efficiency + automation + scalability + monitoring
- examples: data pipelines, model retraining

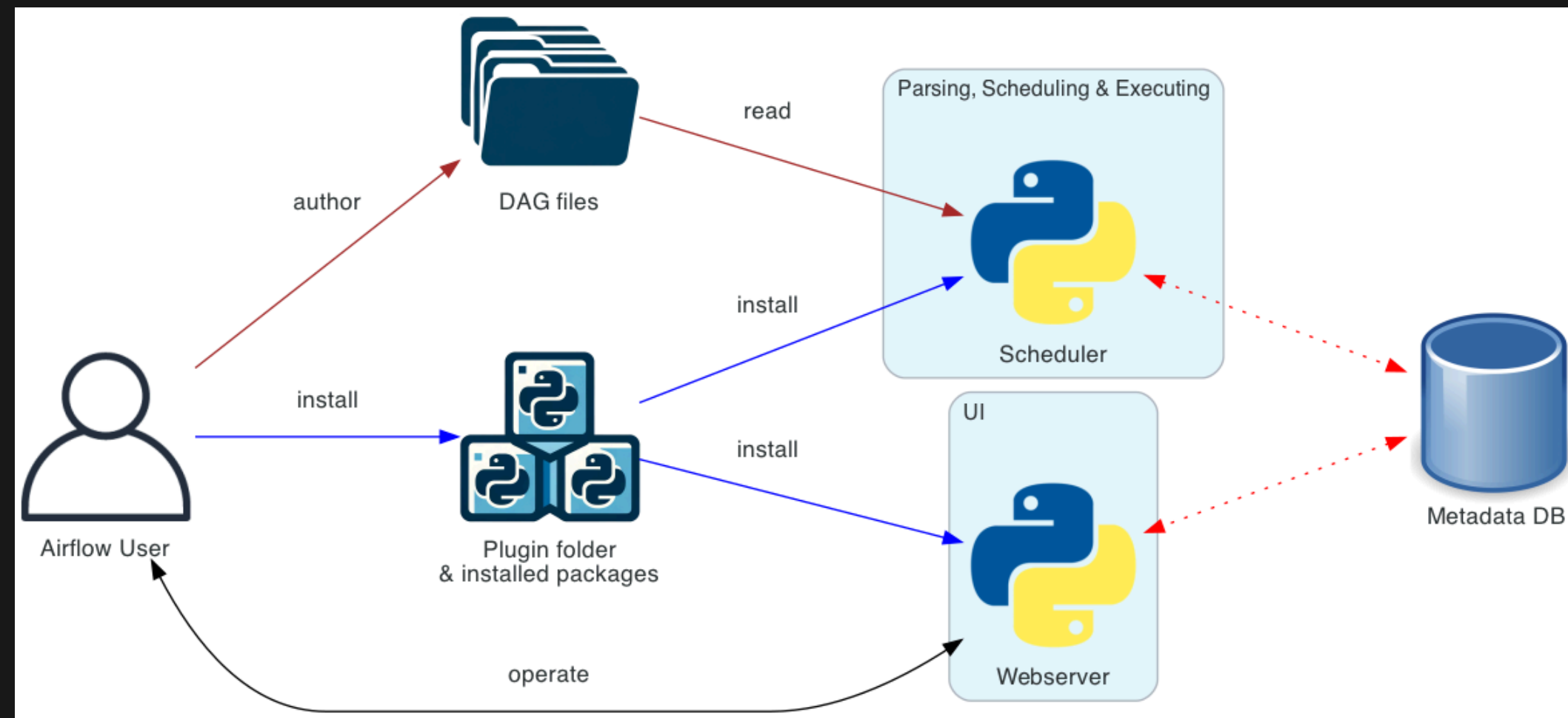
What is Airflow?

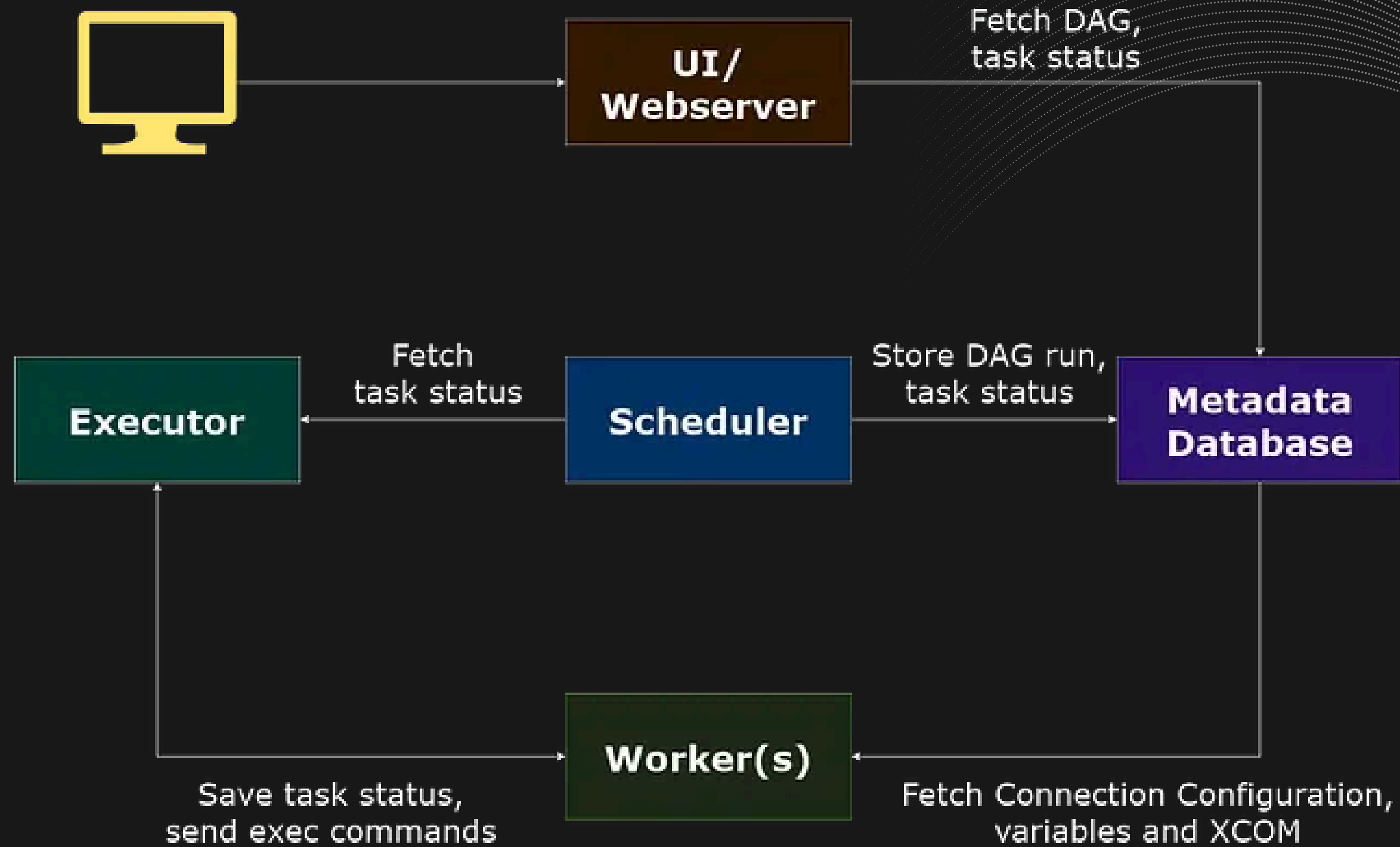
- developing, scheduling, and monitoring workflows
- workflow as (Python) code
- ≥ 1 tasks form a DAG == workflow

Minimal Airflow installation

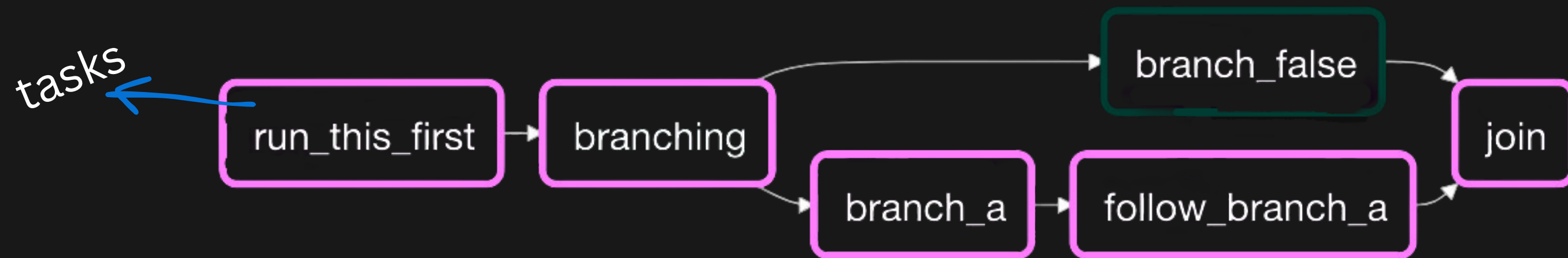
- scheduler
- webserver
- folder of DAG files
- metadata database

Minimal Airflow installation





What is a DAG?

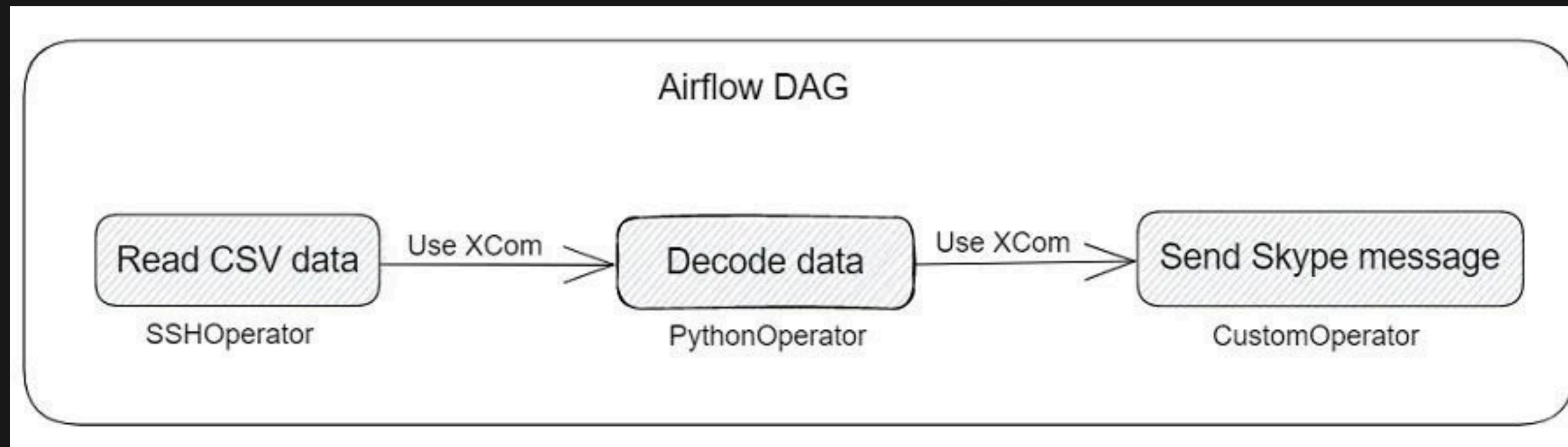


- [python] control flow:
run_this_first >> branching
branching >> branch_a >> follow_branch_a >> join
branching >> branch_false >> join

What is a task?

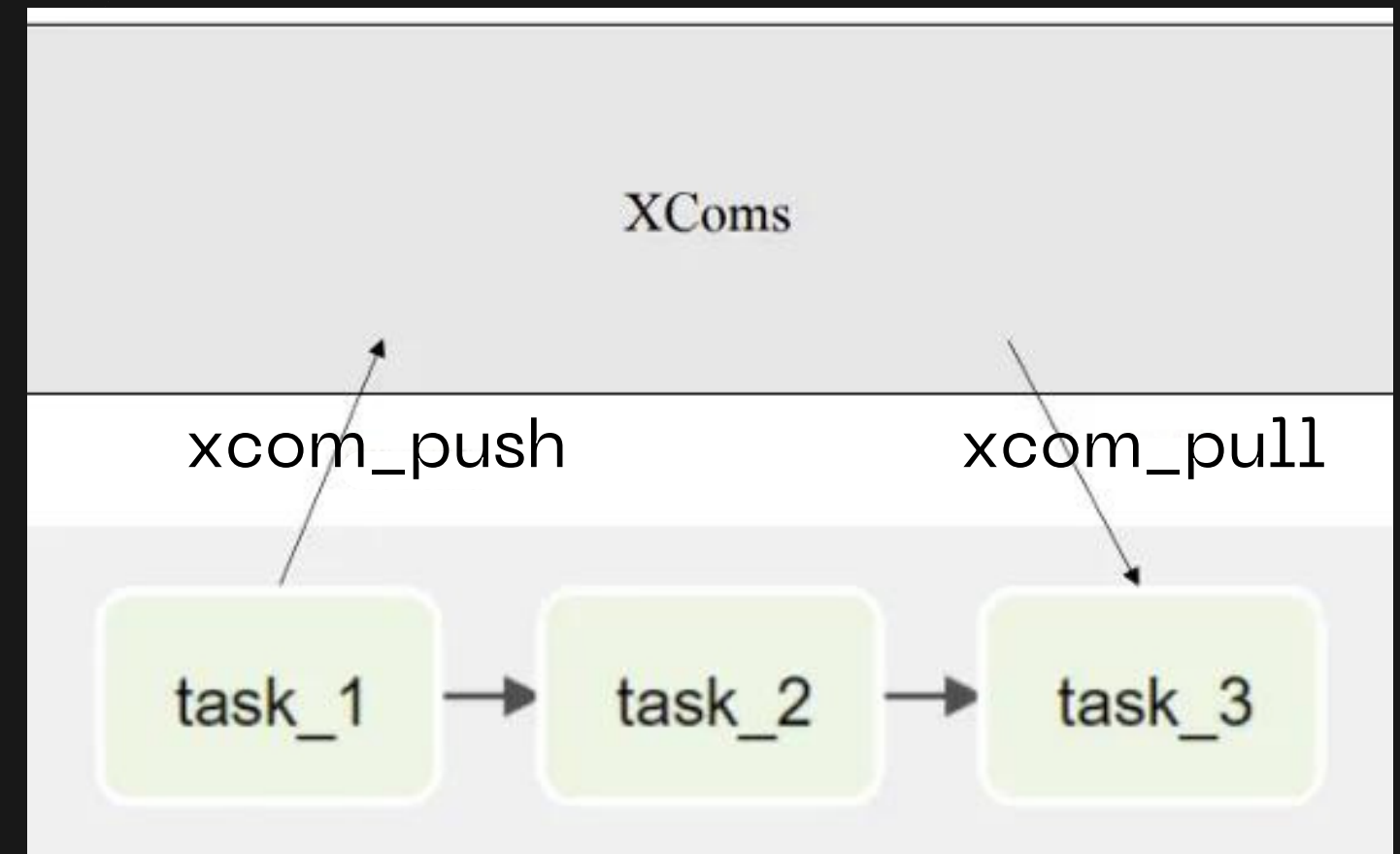
- basic unit of execution
- arranged into DAGs
- Operators: BashOperator, PythonOperator, EmailOperator, PostgresOperator...
- Sensors: wait for something to occur

DAG with Tasks



XComs

- cross-communication mechanism for Tasks
- key, task_id, dag_id
- for small amounts of data
- push vs pull
- inside task_instance of context object



User Interface



DAGsSecurityBrowseAdminDocs

21:11 UTCRH

DAGs

All 26Active 10Paused 16

Filter DAGs by tag

Search DAGs

DAG	Owner	Runs	Schedule	Last Run	Recent Tasks	Actions	Links
example_bash_operator exampleexample2	airflow	2	0 0 ***	2020-10-26, 21:08:11	6		
example_branch_dop_operator_v3 example	airflow		* / 1 * * * *				
example_branch_operator exampleexample2	airflow	1	@daily	2020-10-23, 14:09:17			
example_complex exampleexample2example3	airflow	1 1	None	2020-10-26, 21:08:04	37		
example_external_task_marker_child	airflow	1	None	2020-10-26, 21:07:33			
example_external_task_marker_parent	airflow	1	None	2020-10-26, 21:08:34	1		
example_kubernetes_executor exampleexample2	airflow		None				

Key Takeaways



Orchestrate

Apache Airflow